

Answer Key — Biology Grade 10

Sub-Strand 1.2: Chemicals of Life • Lesson 4: Building Life's Molecules: Structure of Carbohydrates and Proteins

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
C	A	C	A	B	C	D	B

USE AFTER: OBSERVE PHASE — Activity A: Guided Reading & Annotated Diagrams (From Sugar to Starch)

Q1. Ugali is mostly starch, which is a long chain made of many small units joined together. Which small unit is repeated in a starch chain?

- A. Amino acid
- B. Maltose
- C. Glucose**
- D. Glycogen

Correct answer: C. Starch is a polysaccharide built from many glucose monomers joined together. Amino acids are the units of proteins, not starch.

USE AFTER: OBSERVE PHASE — Activity A: Guided Reading & Annotated Diagrams (annotating the bond type)

Q2. Two glucose molecules join to form maltose, a disaccharide. What is the name of the bond that joins them?

- A. Glycosidic bond**
- B. Peptide bond
- C. Protein bond
- D. Amino bond

Correct answer: A. Sugar units are joined by glycosidic bonds. Peptide bonds join amino acids, and learners often confuse the two.

USE AFTER: OBSERVE PHASE — Activity B: Model Building (dipeptide and tripeptide)

Q3. In a model, each bottle top is an amino acid and each straw is a peptide bond. A group joins 3 bottle tops in a single chain to make a tripeptide. How many straws (peptide bonds) do they need?

- A. 1
- B. 3
- C. 2**
- D. 4

Correct answer: C. Each peptide bond joins two neighbouring amino acids, so a chain of 3 needs $3 - 1 = 2$ bonds.

USE AFTER: OBSERVE PHASE — Activity B: Model Building (one sentence: why are proteins so varied?)

Q4. Groups built protein models using bottle tops of different colours for different R-groups. What does this model show about why proteins are so varied?

- A. Different amino acids can be joined in many different orders and combinations**
- B. Proteins are made of only one type of amino acid repeated many times
- C. Each protein uses a different kind of bond to join its units
- D. Proteins are varied because they contain more glucose than starch does

Correct answer: A. Amino acids differ in their R-groups and can be linked in any order and length, so many different proteins are possible. Proteins are joined by peptide bonds, not by different bond types.

USE AFTER: OBSERVE PHASE — Activity C: Kenyan Food Molecule Mapping

Q5. Glycogen is stored in animal tissue such as the meat in nyama choma. On a chart with the columns Monosaccharide, Disaccharide, Polysaccharide and Protein, where does glycogen belong?

- A. Monosaccharide
- B. Polysaccharide**
- C. Protein
- D. Disaccharide

Correct answer: B. Glycogen is a long branched chain of many glucose units, so it is a polysaccharide. It is found in meat but is not a protein.

USE AFTER: EXPLAIN PHASE — Whole-class sensemaking discussion (Q2: extra element in amino acids)

Q6. Glucose contains carbon, hydrogen and oxygen only. Which extra element is found in amino acids but NOT in glucose?

- A. Oxygen
- B. Hydrogen
- C. Nitrogen**
- D. Carbon

Correct answer: C. Amino acids contain nitrogen as well as C, H and O. This is the key elemental difference between carbohydrates and proteins.

USE AFTER: EXPLAIN PHASE — Whole-class sensemaking discussion (Sensemaking Summary: why ugali alone leads to kwashiorkor)

Q7. A child from Kisumu eats only ugali every day and develops a swollen belly and thin limbs. Which explanation is best?

- A. Ugali contains no glucose, so the child has no energy at all
- B. Githeri contains more starch than ugali, so it prevents kwashiorkor
- C. Ugali is not enough in quantity, and the child would be healthy if given more ugali
- D. Ugali gives glucose for energy but no amino acids, so the body cannot build proteins such as muscle and albumin**

Correct answer: D. Ugali is starch (glucose units) and gives energy, but it has no amino acids. Without amino acids the body cannot make muscle protein or albumin, which leads to thin limbs and a swollen belly.

USE AFTER: END OF LESSON — End of lesson: after the Phenomenon Connection sentence in the cumulative model (Layer 4: Molecular Structure)

Q8. Which pair correctly matches a molecule with its main function in the body?

- A. Haemoglobin: energy storage
- B. Haemoglobin: a transport protein that carries oxygen**
- C. Glucose: transports oxygen in the blood
- D. Glycogen: body building and repair

Correct answer: B. Haemoglobin is a protein that carries oxygen in the blood. Glycogen stores energy, and glucose is a monomer that releases energy.